Digital Asset Cryptography

Part 5 - Custody & Wallets: Institutional Custody & Exchanges

Compiled by the Spaced Research Ledger

Last updated: 2026-08-31

The institutions holding other people's bitcoin cannot wait for the protocol debates in Part 3 to resolve. Their clients, insurers and regulators are starting to ask for dated post-quantum plans now. This part covers what the custodians have actually done, which is less than the press releases suggest and more than nothing.

One custodian, BitGo, has shipped working tools. The largest, Coinbase, has convened an expert board and set itself a deadline. The rest have published postures. Around them, a diligence apparatus is forming: government task forces, readiness questionnaires, and insurance underwriters pricing quantum planning into premiums.

Section 1 covers BitGo, the furthest along. Section 2 covers Coinbase. Section 3 covers Fireblocks and the wider provider field. Section 4 covers the concentration of ETF custody. Section 5 covers the regulators, questionnaires and pilots applying the pressure.

1. BitGo

The first regulated custodian to ship quantum-risk tooling to clients, and the first to simulate a post-quantum transaction with its core signing technology.

Recent Developments

BitGo launched four Bitcoin quantum-risk tools on 9 July 2026.[1] They are a Quantum Risk Score, a Fix Exposed Addresses workflow, a coin-selection method that prioritises coins by address risk, and updated default address-type controls.[1] The migration workflow guides clients through moving funds from elevated-risk addresses into new addresses with stronger key hygiene. That is address-reuse remediation, the category dominating the exposure numbers in Part 2.[1]

The tools have clear limits. They do not cover Taproot or P2PK addresses, which expose public keys from creation, and BitGo frames the tooling as a bridge pending protocol-level upgrades rather than a permanent solution.[1] No client adoption metrics have been disclosed, and adoption is the variable that determines whether this work matters.[1]

Current State

BitGo and Silence Laboratories announced on 26 May 2026 the completion of what they describe as the first post-quantum MPC transaction simulation by a regulated custodian.[2] MPC, or multi-party computation, splits a signing key so no single machine ever holds it, and is the technology most institutional custody is built on. The simulation used Silence Laboratories' protocol built on ML-DSA, the NIST-standardised post-quantum signature. It was a simulation rather than production, and no specific blockchain was named.[2]

BitGo's chief executive Mike Belshe framed the move as a planning-horizon shift, saying quantum computing has moved from theoretical discussion to an infrastructure planning priority for the industry.[2] No production rollout date was given. The companies said only that they plan to continue developing and testing with select customers.[2]

2. Coinbase

The largest custodian has an expert board, a self-imposed deadline, and cold wallets that its own report counts among the exposed.

Current State

Coinbase plans to deliver an automated quantum-safe signing pipeline within a year of July 2026, using secure enclaves and threshold cryptography, while coordinating with Bitcoin core developers on migration.[3] Its Independent Advisory Board on Quantum Computing and Blockchain published its report on 13 June 2026.[4] Named participants include Yehuda Lindell, Dan Boneh, Scott Aaronson, Justin Drake, Sreeram Kannan and Dahlia Malkhi.[4]

The report's most pointed observation concerns exchanges themselves. Of the roughly 5 million BTC exposed through address reuse, large amounts sit in the cold wallets of known exchanges, though no exchange is named.[4] The board distinguishes those reuse-exposed active funds from the 1.7 million BTC in Satoshi-era addresses, because active funds can be moved by their owners today.[4] Its two recommendations are to begin technical migration work immediately without waiting for the governance debate, and to communicate more clearly on timelines. It declines to endorse any specific solution.[4]

Coinbase's documented cold-storage practice already includes key hygiene relevant to quantum exposure. Vault keys are single-use or reusable, depending on whether the private key was ever fully reconstituted online.[5] Remaining funds move to a new key after reconstitution.[5] Coinbase may also move assets between cold addresses for security maintenance, without client consent.[5]

3. Fireblocks and the Wider Provider Field

Most custody technology providers have published positions rather than products. The clearest pattern is stated priority without a named roadmap.

Current State

Fireblocks' chief executive Michael Shaulov describes migrating Bitcoin to post-quantum signatures as mostly a coordination issue rather than a technical challenge.[6] The company states that post-quantum cryptography is a strategic priority and that it is researching MPC protocol adaptations for the NIST signature schemes.[7] Its public position is that the threat is real but distant, likely over a decade away absent a material shift in progress.[8] It organises its preparation around cryptographic agility, threat modelling and standards engagement, and recommends the industry target quantum readiness by 2030.[8] Its guidance to institutions is framed as questions to understand rather than concrete steps, and it has not disclosed a roadmap with named algorithms for its own platform.[8]

The provider field is wider than the famous names. The institutional digital-asset security platform market in 2026 is led by Fireblocks, BitGo, Anchorage Digital, DFNS, Copper, Ripple Custody, Ledger Enterprise, Fordefi, Safe and Hex Trust.[9] Copper combines MPC custody with optical air-gapping and off-exchange settlement.[9] It has no active custody licence in the US or UK, and faced a July 2026 deadline for European authorisation.[9] That is a licensing constraint rather than a quantum one, but material to counterparty assessment.[9]

On the supply side, Silence Laboratories, BitGo's post-quantum partner, has launched a quantum-safe vault product of its own, meaning custodian capability is beginning to productise independently of the custodians.[10]

4. ETF Custody Concentration

Most spot Bitcoin ETFs keep their coins with one custodian. That turns one company's quantum posture into a systemic question.

Current State

BlackRock's iShares Bitcoin Trust held 734,762 BTC, worth about $48 billion, as of 15 July 2026.[11] BlackRock broadened the fund's quantum-computing risk disclosure in a May 2025 prospectus revision, expanding a brief mention into a detailed section on how quantum advances could undermine digital-asset security. The filing notes that implementing defences would need broad consensus across a decentralised network.[12]

The custody itself concentrates. BlackRock appointed Coinbase as custodian for the trust and added Anchorage as a second custodian in April 2025 as the fund grew.[13] Coinbase serves as custodian and prime execution agent for several competing bitcoin funds as well, concentrating custody for most spot ETFs in one provider.[14] Morgan Stanley extended the pattern, selecting Coinbase and BNY as custodians for its planned bitcoin product.[15]

5. Regulators, Questionnaires and Pilots

Nobody has ordered a custodian to go post-quantum. What exists instead is a thickening mesh of task forces, self-assessments and insurance pricing.

Recent Developments

The US Treasury launched a Quantum-Readiness Task Force on 24 August 2026, with a dedicated digital-assets workstream to coordinate with crypto custodians and infrastructure providers on migration.[16] Singapore's Cyber Security Agency released a Quantum-Safe Migration Handbook and a readiness self-assessment questionnaire in July 2026.[17] And a cross-regional pilot launched in August 2026 is testing quantum-resistant wallets and transfers, using MPC with ML-DSA signatures on a test network.[18] It involves Bison Bank and DK Bank, with regulators from Abu Dhabi, Bhutan and Malta observing.[18]

Current State

In the US, proposed regulatory frameworks for digital-asset custody reference the NIST post-quantum algorithms and government compatibility requirements as security considerations. That is descriptive rather than prescriptive: no named SEC rule or deadline for digital-asset custodians was found.[19] The G7 Cyber Expert Group's January 2026 financial-sector roadmap places 2026 and 2027 as the awareness and planning phase, with cryptographic inventory work in 2027 and 2028.[20]

The commercial pressure is further along than the regulatory kind. Cyber-insurance underwriters are increasingly incorporating quantum readiness into risk questionnaires at policy renewal, and organisations without a coherent transition plan may face higher premiums. That is the clearest documented pressure mechanism found in this material.[19]

One gap is worth recording. The Alternative Investment Management Association publishes a due-diligence questionnaire for digital-asset funds covering custody.[21] No evidence was found that it currently contains a post-quantum section.[21]

About This Report

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